

# PERMUTATIONS AND COMBINATIONS

①

Permutation: A permutation of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of  $r$ -elements of a set is called an  $r$ -permutation.

Permutation is a process of objects selection and arrangements. It is denoted by  $P(n, r)$  or  $nPr$  or  $P_{n,r}$ .

Theorem: If  $n$  and  $r$  are integers with  $0 \leq r \leq n$  then  $P(n, r) = nPr = \frac{n!}{n-r!}$

Combination: while permutation is concerned with arrangement of objects of a particular set, combination deals with selection only. The order of objects has no importance when determining a combination.

An unordered selection of  $r$  elements from a set is called an  $r$ -combination. An  $r$ -combination is simply a subset with  $r$ -elements of the set.

The number of  $r$ -combinations of a set with  $n$  distinct elements is denoted by  $nCr$  or  $C(n, r)$  or  $\binom{n}{r}$

Theorem:  $nCr = \frac{n!}{r!(n-r)!} = \frac{n!}{r!(n-r)!}$  Note:  $P(n, n) = n!$   
 $P(n, 0) = 1$

Ex: Find the values of (i)  $P(6, 4)$  (ii)  $P(8, 6)$  (iii)  $P(7, 7)$  (iv)  $C(5, 2)$   
(v)  $C(7, 0)$  (vi)  $C(8, 2)$ .

Sol: w.k.T  $P(n, r) = \frac{n!}{n-r!}$   $C(n, r) = \frac{n!}{r!(n-r)!}$   $\binom{10}{0} = 1$   
 $\binom{10}{0} = 1$

$$(i) P(6, 4) = \frac{6!}{6-4!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2} = 6 \times 5 \times 4 \times 3 = 30 \times 12 = 360.$$

$$(ii) C(8, 2) = \frac{8!}{2!(8-2)!} = \frac{8 \times 7 \times 6 \times 5 \times 3 \times 2 \times 1}{2 \times 1 \times 6} = 28.$$

Ex List out all the permutations of  $\{a, b, c\}$

All possible permutations of the objects of set  $\{a, b, c\}$  may be written as  $abc, acb, bac, bca, cba, cab$ .

## PERMUTATIONS

$$(i) P(n, n) = n! \quad (ii) P(n, 0) = 1$$

(iii) suppose it is required to find the number (3) of permutations that can be formed from a collection of  $n$  objects of which  $n_1$  are of one type,  $n_2$  are of a second type,  $\dots$ ,  $n_k$  are of  $k^{\text{th}}$  type with  $n_1 + n_2 + \dots + n_k = n$ . Then the number of permutations of the  $n$  objects is

$$\frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

Ex:1 How many different strings (sequences) of length 4 can be formed using the letters of the word FLOWER?

Sol:- The given word has 6 letters all of which are distinct. Therefore the required number of strings is

$$P(6, 4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 360$$

Ex:2 Find the number of permutations of the (4) letters of the word SUCCESS.

Sol:- Totally 7 letters, of which 3 are S; 2 are C and 1 each are U and E. Therefore the required number of permutations is

$$\frac{7!}{3! \cdot 2! \cdot 1! \cdot 1!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1) \times (2 \times 1) \times 1 \times 1}$$

Ex: Find the value of  $n$  so that  $2P(n, 2) + 50 = P(2n, 2)$

Sol:-

Given

$$2P(n, 2) + 50 = P(2n, 2)$$

$$2[n(n-1)] + 50 = \frac{2n!}{(2n-2)!} = \frac{2n(2n-1)(2n-2)!}{(2n-2)!}$$

Results

$$(1) P(n, 0) = nP_0 = \frac{n!}{(n-0)!} = 1$$

$$(2) P(n, 1) = nP_1 = \frac{n!}{(n-1)!} = \frac{n(n-1)!}{(n-1)!} = n$$

$$(3) P(n, 2) = nP_2 = \frac{n!}{(n-2)!} = \frac{n(n-1)(n-2)!}{(n-2)!} = n(n-1)$$

$$2[n^2 - n] + 50 = 2n(2n - 1) \quad (5)$$

$$2n^2 - 2n + 50 = 4n^2 - 2n$$

$$2n^2 + 50 = 4n^2$$

$$50 = 4n^2 - 2n^2$$

$$50 = 2n^2$$

$$25 = n^2$$

$$n = \pm 5$$

Since  $n$  must be positive  $n = 5$ .

\_\_\_\_\_ X \_\_\_\_\_.

Ex: P.T for all integers  $n, r > 0$  if  $n+1 > r$ , then

$$P(n+1, r) = \left( \frac{n+1}{n+1-r} \right) P(n, r).$$

Sol:- W.K.T  $P(n, r) = \frac{n!}{(n-r)!}$  &  $P(n+1, r) = \frac{(n+1)!}{(n+1-r)!}$

$$\therefore \frac{P(n+1, r)}{P(n, r)} = \frac{(n+1)!}{(n+1-r)!} \times \frac{(n-r)!}{n!}$$

$$= \frac{(n+1)!}{n!} \times \frac{(n-r)!}{(n+1-r)!} \quad (6)$$

$5! = 5 \times 4!$

$$= \frac{(n+1) n!}{n!} \times \frac{(n-r)!}{(n+1-r)(n-r)!}$$

$(n+1-r-1)!$

$$\frac{P(n+1, r)}{P(n, r)} = \frac{n+1}{n+1-r}$$

$$\therefore P(n+1, r) = \left( \frac{n+1}{n+1-r} \right) P(n, r).$$

\_\_\_\_\_ X \_\_\_\_\_.

H.W Find the number of distinguishable permutations of the following words:

(1) BASIC (2) PASCAL (3) BANANA (4) PEPPER

(5) CALCULUS (6) DISCRETE (7) STRUCTURES

(8) ENGINEERING (9) MATHEMATICS.

\_\_\_\_\_ X \_\_\_\_\_.

## Combinations

The total number of combinations of  $r$  different objects that can be selected from  $n$  different objects can be obtained by proceeding in the following way. Suppose this number is equal to  $C$ , say, that is suppose there is a total of  $C$  number of combinations of  $r$  different objects chosen from  $n$  different objects. Take any one of these combinations. The  $r$  objects in this combination can be arranged in  $r$  different ways, since there are  $C$  combinations, the total number of permutations is  $\frac{P(n, r)}{r} = C$ . This number is denoted by  $nC_r$  (or)  $c(n, r)$  (or)  $\binom{n}{r}$

$$nC_r = \frac{n!}{(n-r)! r!}$$

PART-A

1  
2  
3  
4

PART-B

1  
2  
3  
4

Ex:1

A certain question paper contains two ⑧  
Parts A and B each containing 4 questions. How many different ways a student can answer 5 questions by selecting at least 2 questions from each part?

Sol:- The different ways a student can select his 5 questions are

(i) 3 questions from part A and 2 questions from part B. This can be done in  $4C_3 \times 4C_2$

$$= \frac{4!}{(4-3)! 3!} \times \frac{4!}{(4-2)! 2!}$$

$$= \frac{4 \times 3!}{1! 3!} \times \frac{4 \times 3 \times 2 \times 1}{2! 2!} = 4 \times 6 = 24 \text{ ways.}$$

(ii) 2 questions from part A and 3 questions from part B. This can be done in  $4C_2 \times 4C_3 = 24$  ways

9

$$24 + 24 = 48 \text{ ways.}$$

Ex: A certain question paper contains three parts A, B, C with four questions in part A, five questions in part B and six questions in part C. It is required to answer seven questions selecting at least two questions from each part. In how many different ways can a student select his seven questions for answering?

Sol:- The different possible ways in which a student can make a selection are

- Can make:
- (i) 2 questions from part A, 2 from part B and 3 from part C.
  - (ii) 2 questions from part A, 3 from part B and 2 from part C.

- (iii) 3 questions from Part A, 2 from Part B and 2 from Part C. (10)

the selection (I) can be made in

$${}^4C_2 \times {}^5C_2 \times {}^6C_3 = 6 \times 10 \times 20 = 1200 \text{ ways.}$$

The selection (II) can be made in

$$4C_2 \times 5C_3 \times 6C_2 = 6 \times 10 \times 15 = 900 \text{ ways.}$$

and the selection (III) can be made in

$$4C_3 \times 5C_2 \times 6C_2 = 4 \times 10 \times 15 = 600 \text{ ways.}$$

$\therefore$  Total no. of ways a student can answer 7 questions under the given restrictions is

$$1200 + 900 + 600 = 2700 \text{ ways.}$$

| <u>1947-8</u> | <u>1948-9</u> | <u>1949-50</u> | <u>1950-1</u> |
|---------------|---------------|----------------|---------------|
| 1             | 1             | 1              | 1             |
| 2             | 2             | 2              | 2             |
| 3             | 3             | 3              | 3             |
| 4             | 4             | 4              | 4             |
| 5             | 5             | 5              | 5             |

Note:-

(11)

$$* nC_n = nC_0 = 1$$

$$* nC_1 = nC_{n-1} = n$$

For  $r > n$ ,  $nC_r$  is defined to be equal to zero.

Properties of combinations:

$$1. nC_0 = nC_n = 1$$

$$2. nC_r = nC_{n-r}$$

$$3. nC_r + nC_{r-1} = n+1C_r$$

$$4. nC_r = \frac{n}{r} n-1C_{r-1}$$

$$5. n n-1C_{r-1} = (n-r+1) nC_{r-1}$$

$$\begin{aligned} 100C_{97} &= \frac{100}{100 \times 97 \times 96} \\ &= \frac{100}{3 \times 97 \times 96} \\ &= \frac{100}{28224} \\ &= \frac{1}{282} \end{aligned}$$

Note The result  $nC_r = nC_{n-r}$

For example, find the value of  $100C_{97}$

$$100C_{97} = 100C_{100-97} = 100C_3 = \frac{100 \times 99 \times 98}{3 \times 2 \times 1} =$$

(1) Prove that  $n+1C_r = nC_{r-1} + nC_r$ .

(12)

$$C(n+1, r) = C(n, r-1) + C(n, r)$$

proof

$$\text{Given } C(n, r-1) + C(n, r) = \frac{n!}{(n-r+1)!(r-1)!} + \frac{n!}{(n-r)!r!}$$

$$= \frac{n!}{(r-1)!(n-r)!} \left\{ \frac{1}{n-r+1} + \frac{1}{r} \right\}$$

$$(\because r = r(r-1)!) \quad \text{--- (1)}$$

$$= \frac{n!}{(r-1)!(n-r)!} \left\{ \frac{r+n-r+1}{r(n-r+1)} \right\}$$

$$\begin{aligned} (\because (n-r+1)! &= (n-r+1)(n-r)! \\ &= (n-r+1) \cdot (n-r)! \end{aligned}$$

$$= \frac{n!}{(r-1)!(n-r)!} \left\{ \frac{n+1}{r(n-r+1)} \right\}$$

$$\begin{aligned} [\because r &= r(r-1)!] \\ [\because n(n+1) &= (n+1)!] \end{aligned}$$

$$= \frac{(n+1)!}{r!(n-r+1)!}$$

$$= C(n+1, r)$$

$$(n+1)! = (n+1)n!$$

$$\begin{aligned} (n+1)! &= (n+1)n! \\ (n+1)! &= (n+1)n! \\ (n+1)! &= (n+1)n! \end{aligned}$$

② There are 7 men and 3 women. (13)

Find the number of ways in which a Committee of 6 persons can be formed if the committee is to have

(a) exactly 4 men and

(b) at least 2 women.

Sol:-

(a) A committee consisting exactly 4 men can be formed in

$${}^7C_4 \times {}^3C_2 = \frac{7!}{4!3!} \times \frac{3!}{2!1!}$$

$$= \frac{7 \times 6 \times 5 \times 4!}{4!3!} \times \frac{3 \times 2 \times 1}{2 \times 1}$$

$$= 35 \times 3$$

$$= \overset{210}{\cancel{105}} \text{ ways.}$$

(b) The committee consisting at least 2 women can be formed as follows:

(i) 4 men and 2 women (ie)  ${}^7C_4 \times {}^3C_2$

$$= \overset{210}{\cancel{105}} \text{ ways.}$$

(ii) 3 men and 3 women (ie)  ${}^7C_3 \times {}^3C_3 = 35 \text{ ways.}$

~~(iii) 2 men and 4 women (ie)  ${}^7C_2 \times {}^3C_4 = 21 \text{ ways.}$~~

$\overset{105}{\cancel{35}}$   
 $\underline{140}$

5M  
6Fm.  
Committee of 5 is  
to be formed.

April 2020

(14)

③ Out of 5 males and 6 females, a committee of 5 is to be formed. Find the number of ways in which it can be formed so that among the persons chosen in the committee there are:

(a) exactly 3 males and 2 females

(b) at least 2 males and one female.

Sol:-

(a) The total number of ways in which the committee can be formed is

$$\begin{aligned}
 {}^5C_3 \times {}^6C_2 &= \frac{5!}{\underline{5-3} \ 3} \times \frac{6!}{\underline{6-2} \ 2} \\
 &= \frac{5!}{2! \ 3!} \times \frac{6!}{4! \ 2!} \\
 &= 10 \times 15 \\
 &= 150.
 \end{aligned}$$

(b) A committee of 5 members with exactly 3 males and 2 females can be formed by selecting manner:

(i) 4 males and 1 females in  ${}^5C_4 \times {}^6C_1$  ways

(ii) 3 males and 2 females in  ${}^5C_3 \times {}^6C_2$  ways

(15)

(iii) 2 males and 3 females in

 $5C_2 \times 6C_3$  ways.

Hence the total no. of different ways to form a committee of 5 is

$$\begin{aligned}
 & 5C_4 \times 6C_1 + 5C_3 \times 6C_2 + 5C_2 \times 6C_3 \\
 &= \frac{5!}{4!1!} \times \frac{6}{1511} * \frac{15}{1312} \times \frac{16}{1214} + \frac{15}{1312} \times \frac{16}{1313} \\
 &= 30 + 150 + 200 \\
 &= 380.
 \end{aligned}$$

\_\_\_\_\_ x \_\_\_\_\_.

(4) How many triangles can be made by joining 12 points in a plane, given that 7 are in one line?

Sol:-



Since a triangle can only be formed by joining 3 points out of 12 points in a plane. Thus the total number of triangles can be formed in  $12C_3$  ways.

(16)

But 7 points are in one line,  $\therefore$  out of these 7 points, a triangle can be formed with 3 points. Thus the total number of triangles can be formed in  ${}^7C_3$  ways. Hence the required number of triangles are

$$\begin{aligned}
 {}^{12}C_3 - {}^7C_3 &= \frac{12!}{(12-3)!3!} - \frac{7!}{(7-3)!3!} \\
 &= \frac{12 \cdot 11 \cdot 10 \cdot 9}{9 \cdot 3 \times 2 \times 1} - \frac{7 \cdot 6 \cdot 5 \cdot 4}{4 \cdot 3 \times 2 \times 1} \\
 &= 220 - 35 \\
 &= 185.
 \end{aligned}$$

—————X—————

(5) If  ${}^nC_x = 56$  and  ${}^nP_x = 336$  find  $n$  and  $x$ .

Sol:- Dividing  ${}^nP_x$  by  ${}^nC_x$ , we get

|                                                                                                             |                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{{}^nP_x}{{}^nC_x} = \frac{336}{56}$ $\frac{\frac{n!}{(n-x)!}}{\frac{n!}{(n-x)!x!}} = \frac{336}{56}$ | $1 \times = 6 = 13$ $\boxed{x=3}$ <p>Substituting value of <math>x</math> in <math>{}^nC_x</math></p> <p>we set</p> ${}^nC_3 = 56$ $\frac{n!}{(n-3)!3!} = 56$ |
|                                                                                                             | $\frac{n(n-1)(n-2)}{3!} = 56$ $n^3 - 3n^2 + 2n - 336 = 0$ $\boxed{n=8}$                                                                                       |

PERMUTATION WITH REPETITION (17)

Counting permutations when repetition of elements is allowed can easily be done using the product rule.

Ex: How many strings of length  $r$  can be formed from the English alphabet?

Soln:

By product rule, because there are 26 letters and because each letter can be used repeatedly,

$\therefore$  There are  $26^r$  strings of length  $r$ . 

Note:-

The no. of  $r$ -Permutations of a set of  $n$  objects with repetition allowed is  $n^r$ .

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Combinations with Repetitions

Suppose to select with repetition a combination of  $r$  objects from a set of  $n$  distinct objects. The no. of such selections is given by

(18)

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r! (n-1)!} = \binom{r+n-1}{n-1}$$

$$C(n+r-1, r) = C(r+n-1, n-1)$$

$\therefore C(n+r-1, r) = C(r+n-1, n-1)$  represents the no. of combinations of  $n$  objects, taken  $r$  at a time, with repetitions allowed.

The following are other interpretations of this number

(i)  $C(n+r-1, r) = C(r+n-1, n-1)$  represents the no. of ways in which  $r$  identical objects can be distributed among  $n$  distinct containers.

(ii)  $C(n+r-1, r) = C(r+n-1, n-1)$  represents the number of non-negative integer solutions of the equation,

$$x_1 + x_2 + \dots + x_n = r$$

Ex:1 A bag contains coins of seven different denominations, with at least one

dozen coins in each denomination. In (19) how many ways can we select a dozen coins from the bag?

Sol:

The selection consists in choosing with repetitions  $r=12$  coins of  $n=7$  distinct denominations. The number of ways of making this selection is

$$C(n+r-1, r) = C(7+12-1, 12)$$

$$= C(18, 12)$$

$$= {}^{18}C_{12}$$

$$= 18,564.$$

$$\therefore n C_r = \frac{n!}{n-r!}$$

————— X —————

(2) In how many ways can we distribute  $r=7$  apples and  $r=6$  oranges among  $n=4$  children so that each child gets at least 1 apple?

Sol:

Suppose we give 1 apple to each child

$\therefore$  4 apples exhausts.

[  $7-4=3$  ]  
apples.

(20)

The remaining 3 apples can be distributed among the 4 children in

$$C(4+3-1, 3) = C(6, 3)$$

Also 6 oranges can be distributed among 4 children in

$$C(4+6-1, 6) = C(9, 6) \text{ ways.}$$

=  ${}^9C_6$ 

∴ By Product Rule

The no. of ways of distributing the given fruits under the given condition is

$$C(6, 3) \times C(9, 6) = 1680.$$

$${}^6C_3 \times {}^9C_6 =$$

$$= {}^6C_3$$

$$= \frac{16}{1}$$

$$\frac{16 \times 13}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= \frac{16 \times 13}{3 \times 2 \times 1 \times 3 \times 2 \times 1}$$

$$= \frac{8 \times 13}{1 \times 1 \times 1 \times 1 \times 1 \times 1}$$

8A

$$= 20 \text{ ways.}$$

3 Sunday

(2) Find the no. of non-negative integer solution of the equation  $x_1 + x_2 + x_3 + x_4 + x_5 = 8$ .

Sol:

Here  $n=5$ ,  $r=8$ .

W.K.T

$$C(n+r-1, r) = C(5+8-1, 8)$$

$$= C(12, 8)$$

$$= {}^{12}C_8$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10 \times 11 \times 12} = 495.$$

————— X —————

④ Find the no. of non-negative integer solutions of the inequality

$$x_1 + x_2 + x_3 + \dots + x_6 < 10.$$

Sol:-

We have to find the no. of non-negative integer solutions of the equation

$$x_1 + x_2 + x_3 + \dots + x_6 = 9 - x_7$$

where  $9 - x_7 \leq 9$  so that  $x_7$  is a non-negative integer.  $\left[ \begin{array}{l} -x_7 \leq 9 - 9 \leq 0 \\ x_7 \geq 0 \end{array} \right]$

$\therefore$  The required number is the number of non-negative solutions of the equation

$$x_1 + x_2 + \dots + x_7 = 9$$

This number is

Hence

$$\begin{array}{l} n=7 \\ r=9 \end{array}$$

$$C(n+r-1, r) = C(7+9-1, 9)$$

$$= C(15, 9)$$

$$= {}^{15}C_9$$

$$= \frac{{}^{15}P_9}{15-9}$$

$$= \frac{115}{16 \cdot 19} = 5005.$$

\_\_\_\_\_ X \_\_\_\_\_

$$\begin{array}{l} -x_7 \leq 9-9 \\ -x_7 \leq 0 \\ x_7 \geq 0 \end{array}$$

⑤ Find the number of Positive integer (22)  
 Solutions of the equation  $x_1 + x_2 + x_3 = 17$ .

Sol:-

Here we require  $x_1 \geq 1$ ,  $x_2 \geq 1$ ,  $x_3 \geq 1$ .

Let us set  $y_1 = x_1 - 1$ ,  $y_2 = x_2 - 1$ ,  $y_3 = x_3 - 1$ .

Then  $y_1, y_2, y_3$  are all non-negative integers

$$(y_1 + 1) + (y_2 + 1) + (y_3 + 1) = 17.$$

$$y_1 + y_2 + y_3 = 17 - 3 = 14.$$

$$\text{Here } n = 3, r = 14.$$

12 Sunday The number of non-negative solutions  
 of this equation is the required number.

Week 16

This number is

$$C(n + r - 1, r) = C(3 + 14 - 1, 14)$$

$$= C(16, 14)$$

$$= {}^{16}C_{14}$$

$$= \frac{16}{16-14} \cdot 14$$

$$= \frac{16}{2} \cdot 14$$

$$= \frac{16 \times 15 \times 14}{2 \times 14}$$

$$= 120.$$

⑥ Find the number of integer solutions (23)

$$\text{of } x_1 + x_2 + x_3 + x_4 + x_5 = 30$$

$$\text{where } x_1 \geq 2, x_2 \geq 3, x_3 \geq 4, x_4 \geq 2, x_5 \geq 0.$$

Soln

$$\text{let } y_1 = x_1 - 2, y_2 = x_2 - 3, y_3 = x_3 - 4,$$

$$y_4 = x_4 - 2, y_5 = x_5.$$

Then  $y_1, y_2, y_3, y_4, y_5$  are all non-negative integers.

$$\therefore (y_1 + 2) + (y_2 + 3) + (y_3 + 4) + (y_4 + 2) + y_5 = 30$$

$$y_1 + y_2 + y_3 + y_4 + y_5 = 30 - 11$$

$$= 19$$

$$\text{Here } n = 5, r = 19.$$

w.k.t

$$C(n+r-1, r) = C(5+19-1, 19)$$

$$= C(23, 19)$$

$$= {}^{23}C_{19}$$

$$\left[ {}^nC_r = \frac{n!}{r!(n-r)!} \right]$$

$$= \frac{23!}{(23-19)! 19!}$$

$$= \frac{23!}{4! 19!}$$

$$\rightarrow 8855$$

BINOMIAL THEOREM

(24)

If  $x$  &  $y$  are variables and  $n$  is a positive integer, then

$$(x+y)^n = \sum_{r=0}^n {}^nC_r x^r y^{n-r}$$

Here the term  ${}^nC_r$  is often referred to as Binomial Coefficient. The B.C is also written as  $C(n, r)$  (or)  $\binom{n}{r}$ .

MULTINOMIAL THEOREM

For positive integers  $n$  and  $t$ , the coefficient of  $x_1^{n_1}, x_2^{n_2}, \dots, x_t^{n_t}$  in the expansion of  $(x_1 + x_2 + \dots + x_t)^n$  is

$$\frac{n!}{n_1! n_2! \dots n_t!}$$

where each  $n_i$  is a non-negative integer  $\leq n$  and  $n_1 + n_2 + \dots + n_t = n$ .

[OR]

The general term in the expansion of

$$(x_1 + x_2 + \dots + x_t)^n = \frac{n!}{n_1! n_2! \dots n_t!} x_1^{n_1} \dots x_t^{n_t}.$$

Where  $n_1, n_2, \dots, n_t$  are non negative 25 integers not exceeding  $n$  and  $n_1 + n_2 + \dots + n_t = n$ .

The expression  $\frac{n!}{n_1! n_2! \dots n_t!}$  is

also written as  $\binom{n}{n_1, n_2, \dots, n_t}$  and

is called the multinomial coefficient.

Ex:1 Prove the following identities for a positive integer  $n$

$$(i) \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

$$(ii) \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0.$$

Sol:-

$$\text{w.k.T } (x+y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$$

When  $x=y=1$  this becomes

$$\begin{aligned} 2^n &= \sum_{r=0}^n \binom{n}{r} (1)^r (1)^{n-r} \\ &= \sum_{r=0}^n \binom{n}{r} = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} \end{aligned}$$

and when  $x = -1$  and  $y = 1$  we get

(26)

$$0 = \sum_{r=0}^n \binom{n}{r} (-1)^r = \binom{n}{0} - \binom{n}{1} + \binom{n}{2}$$

$$\dots + (-1)^n \binom{n}{n}$$

— x —

✓ Ex:2 Find the coefficient of  $x^9 y^3$  in the expansion of  $(2x - 3y)^{12}$ .

Sol:

W.K.T

By Binomial theorem

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$$

Here  $n=12$ ,  $r=9$

Now

$$(2x - 3y)^{12} = \sum_{r=0}^{12} \binom{12}{r} (2x)^r (-3y)^{12-r}$$

$$= \sum_{r=0}^{12} \binom{12}{r} 2^r x^r (-3)^{12-r} (y)^{12-r}$$

$$r=9 \quad = \binom{12}{9} 2^9 (-3)^3$$

The coefficient of  $x^9 y^3$  is

$$\binom{12}{9} (2)^9 (-3)^3 = 2^9 \times (-3)^3 \times {}^{12}C_9$$

$$= 512 \times -27 \times 220$$

$$= -30,41,280$$

Ex: 3. The coefficient of  $x^5 y^2$  in the expansion of  $(x+y)^7$ . (27)

Sol:-

w.k.T Binomial theorem

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$$

Here  $n=7$ ,

Now

$$(x+y)^7 = \sum_{r=0}^7 \binom{7}{r} x^r y^{7-r}$$

$$= \binom{7}{5} (x^5 y^2) + \dots$$

Put  $r=5$

The coefficient of  $x^5 y^{7-5} = x^5 y^2$  is

$$= \binom{7}{5} (1)^5 (1)^2 = \binom{7}{5} = {}^7C_5$$

$$= \frac{7 \times 6 \times 5}{2 \times 5} = \frac{7 \times 6}{2} = 21.$$

————— x —————.

Ex: 4 Evaluate  $\binom{12}{5, 3, 2, 2}$

Sol:- w.k.T  $\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$

Now

$$\binom{12}{5, 3, 2, 2} = \frac{12!}{5! 3! 2! 2!} = 1,66,320. \quad (28)$$

————— X —————

Ex: 5 Find the term which contains  $x^{11}$  and  $y^4$  in the expansion of  $(2x^3 - 3xy^2 + z^2)^6$ .

Sol:-

By the multinomial theorem, the general term in the given expansion is

1 Sunday

$$\binom{6}{n_1, n_2, n_3} = \frac{6!}{n_1! n_2! n_3!} x_1^{n_1} x_2^{n_2} x_3^{n_3}$$

Week 10

To find  $n_1, n_2, n_3$

$$\binom{6}{n_1, n_2, n_3} (2x^3)^{n_1} (-3xy^2)^{n_2} (z^2)^{n_3} = (2x^3 - 3xy^2 + z^2)^6$$

Here  $x_1 = 2x^3$   
 $x_2 = -3xy^2$   
 $x_3 = z^2$

$$= \left( \frac{6!}{n_1! n_2! n_3!} \right) 2^{n_1} (-3)^{n_2} x^{3n_1} x^{n_2} y^{2n_2} z^{2n_3}$$

$$(2x^3 - 3xy^2 + z^2)^6 = \left( \frac{6!}{n_1! n_2! n_3!} \right) 2^{n_1} (-3)^{n_2} x^{3n_1 + n_2} y^{2n_2} z^{2n_3}$$

The term containing  $x^{11}$  &  $y^4$  is

$$11 = 3n_1 + n_2 \rightarrow (1)$$

(29)

$$3n_1 + n_2 = 11 \rightarrow (1)$$

$$2n_2 = 4 \rightarrow (2)$$

$$n_2 = \frac{4}{2} = 2.$$

$$\text{Put } n_2 = 2 \text{ in (1)}$$

$$3n_1 + 2 = 11 \Rightarrow 9 \Rightarrow n_1 = 3.$$

$$\text{Since } n_1 + n_2 + n_3 = 6, \quad 3 + 2 + 1 = 6$$

$$\text{We should have } n_3 = 1, \quad 3 + 2 + n_3 = 6 \Rightarrow n_3 = 6 - 5 = 1$$

$$\therefore n_1 = 3, n_2 = 2, n_3 = 1.$$

According to the term containing  $x''$  and

$y^4$  is

$$\binom{6}{3 \ 2 \ 1} \frac{6!}{3! \ 2! \ 1!} 2^3 (-3)^2 x'' y^4 z^2$$

$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 4} \times 8 \times 9 \times x'' y^4 z^2 \quad 6 \times 5 \times 4 \times$$

$$= 30 \times 72 \times x'' y^4 z^2$$

$$= \frac{6!}{3! \ 2! \ 1!} 2^3 \times (-3)^2 \times x'' y^4 z^2$$

$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 4} \times 8 \times 9 \times x'' y^4 z^2$$

$$= 4320 x'' y^4 z^2 //$$

Ex: 6 Determine the coefficient of (30)

(i)  $x^1 y^1 z^2$  in the expansion of  $(2x - y - z)^4$  and

(ii)  $a^2 b^3 c^2 d^5$  in the expansion of  $(a + 2b - 3c + 2d + 5)^{16}$ .

Sol:-

By the multinomial theorem

$$(2x - y - z)^4 = \left( \frac{4}{n_1 n_2 n_3} \right) (2x)^{n_1} (-y)^{n_2} (-z)^{n_3}$$

$$= \left( \frac{4}{n_1 n_2 n_3} \right) 2^{n_1} x^{n_1} (-1)^{n_2} y^{n_2} (-1)^{n_3} z^{n_3}$$

For  $n_1 = 1, n_2 = 1, n_3 = 2$  [Since  $n_1 + n_2 + n_3 = 4$ ]

$$= \frac{4!}{1! 1! 2!} 2^{(1)} x^1 (-1)^1 (y)^1 (-1)^2 (z)^2$$

$$= \frac{4!}{1! 1! 2!} 2 \times (-1) \times 1 \times (-1)^2 x y z^2$$

$$= \frac{4 \times 3 \times 2 \times 1}{1 \times 1 \times 2 \times 1} (-2) x y z^2$$

$$= -24 x y z^2$$

(31)

By the multinomial theorem,

$$(a + 2b - 3c + 2d + 5)^{16}$$

$$= \binom{16}{n_1 n_2 n_3 n_4 n_5} (a)^{n_1} (2b)^{n_2} (-3c)^{n_3} (2d)^{n_4} (5)^{n_5}$$

$$= \binom{16}{n_1 n_2 n_3 n_4 n_5} (a)^2 (2b)^3 (-3c)^2 (2d)^5 (5)^5$$

$$[\because a^2 b^3 c^2 d^5]$$

$$= \binom{16}{2! 3! 2! 5! 4!}$$

$$\left[ \begin{array}{l} n_1 + n_2 + n_3 + n_4 + n_5 = 16 \\ 2 + 3 + 2 + 5 + n_5 = 16 \\ n_5 = 16 - 12 \\ = 4 \end{array} \right]$$

$$(1)^2 (2)^3 (-3)^2 \times (2)^5 (5)^4 = 4 \quad a^2 b^3 c^2 d^5$$

$$= \frac{16!}{2! 3! 2! 5! 4!}$$

$$= 8 \times 9 \times 32 \times 625 \times \frac{16!}{2! 3! 2! 5! 4!} a^2 b^3 c^2 d^5$$

= Thus, the required coefficient is

$$\frac{16! \times 2^5 \times 5^3 \times 3}{(4!)^2}$$

$$(4!)^2$$

Ex: 7 Find the coefficient of

(32)

(i)  $x^5 y^2$  in the expansion of  $(2x-3y)^7$ .

(ii)  $xyz^{+2}$  in the expansion of  $(x-2y+3z)^4$ .

Soln

(i) By Binomial coefficient

$$(2x-3y)^7 = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$$

$$(2x-3y)^7 = \sum_{r=0}^7 \binom{7}{r} x^r y^{7-r}$$

put  $r=5$

$$= \sum_{r=0}^7 \binom{7}{r} (2x)^r (-3y)^{7-r}$$

$$= \sum_{r=0}^7 \binom{7}{5} (2x)^5 (-3y)^{7-5}$$

$$= \sum_{r=0}^7 \binom{7}{5} (2)^5 x^5 (-3)^2 (y)^2$$

The coefficient of  $x^5 y^2$  is

$$\binom{7}{5} 2^5 (-3)^2 = {}^7C_5 2^5 (-3)^2$$

$$= \frac{7 \times 6 \times 5}{1 \times 2 \times 1} \times 32 \times 9$$

$$= 6048 //$$

(ii) By multinomial coefficient

(33)

W.K.T

 $xyz^2$ 

$$\binom{n}{n_1, n_2, n_3, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}$$

$$= \binom{4}{n_1, n_2, n_3} x_1^{n_1} (-2y)^{n_2} (3z)^{n_3}$$

$$= \binom{4}{1, 1, 2} (1)^1 (-2y)^1 (3)^2 xyz^2$$

Equating the coefficient of  $xyz^2$ 

$$= \frac{4!}{1! 1! 2!} (-2)(3)^2$$

$$= \frac{4 \times 3 \times 2 \times 1}{1 \times 1 \times 2 \times 1} (-2) \times 9$$

$$= 12 \times -18$$

$$= -216$$

$$\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

Note:

(34)

| type                | Repetition | Formula                      |
|---------------------|------------|------------------------------|
| ① $r$ -Permutations | No         | $\frac{n!}{(n-r)!}$          |
| ② $r$ -combinations | No         | $\frac{n!}{(n-r)! r!}$       |
| ③ $r$ -Permutations | yes        | $n^r$                        |
| ④ $r$ -combinations | yes        | $\frac{(n+r-1)!}{r! (n-1)!}$ |

—————X—————

H.W

Find the coefficient of

(i)  $x^3 y^3 z^2$  in the expansion of  $(2x-3y+5z)^8$ .(ii)  $x_1^2 x_3 x_4^3 x_5^4$  in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$ .Evaluate (i)  $\binom{7}{2, 3, 2}$  (ii)  $\binom{10}{5, 3, 2, 2}$ 

—————X—————

Ex. 1.1.1

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

(35)

Find  $n_1, n_2, n_3, n_4, n_5$

$$= \binom{10}{1^2 1^4 1^2 1^4}$$

$x_1^2 x_2^0 x_3^1 x_4^3 x_5^4$

- ① Expand  $(x+2y)^5$  using Binomial theorem
- ② State multinomial theorem and determine the coefficient of  $x^4 y^6$  in  $(3x-4y)^{10}$ .
- ③ Find the coefficient of  $x^8 y^4$  in the expansion of  $(2x+3y)^{12}$
- ④ Find the coeff of  $xyz^2$  in the expansion of  $(2x-y-z)^4$
- ⑤ State Binomial and determine the coeff of  $x^2 y^3$  in the expansion of  $(2x-3y)^{25}$

$n_1=2$   
 $n_2=0$   
 $n_3=1$   
 $n_4=3$   
 $n_5=4$

$n_1+n_2+n_3+n_4+n_5=10$   
 $2+n_2+1+3+4=10$   
 $n_2=0$

$n_1+n_2+n_3+n_4+n_5=10$   
 $2+n_2+1+3+4=10$   
 $n_2=0$

- ⑥ Find the coeff of  $x^9 y^3$  in the expansion of  $(x+y)^{12}$
- ⑦ ...
- ⑧ ...
- ⑨ ...

$x^5 y^2 \dots (x+y)^7$   
 $x^5 y^2 \dots (2x-3y)^7$   
 $xyz^2 \dots (x-2y+3z)^4$

21 Saturday

Week 12

RAMCO

March 2020

Page: 19

① In how many ways can we distribute 12 identical pencils to 5 children so that every child gets at least 1 pencil? (36)

Sol: First we distribute one pencil to each child. Then there remain 7 pencils to be distributed. The number of ways of distributing these 7 pencils to 5 children is the required number.

This number is

$$C(n+r-1, r) = C(5+7-1, 7) \\ = C(11, 7)$$

22 Sunday

Week 13

$$= \frac{11!}{4! 7!} = 330 \text{ ways}$$

note

Sum Rule:

Simultaneous cannot do this work (class or boys or girls)

Product Rule: Simultaneous can do this work (A & B) (both)